

coo4one research findings — Heptabase + Apple Notes

Source labels: [H] = Heptabase card, [H-WB] = Heptabase whiteboard, [H-J <date>] = Heptabase journal entry, [AN: folder] = Apple Note (folder shown). Apple Notes don't carry visible dates here; Heptabase ones do.

1. Dream-system thinking & modular software (the dominant theme)

[H] **“The Case for Modularity: Why Modern Software Feels Broken”** — dream-system spec / frustration with current tools > “My task manager doesn't talk to my notes. My files, locked in an impenetrable cloud ecosystem, refuse to cooperate with the apps I need. It's not that these tools are bad; it's that they feel indifferent to me, their user... the tools we use to think, create, and organize our lives have become a barrier instead of an enabler.”

[H] **“We need more modular software”** — dream-system spec (cross-tags with PKM frustration) > “Most software contains three elements - data (content), data control (query, modification, organization), and display. In an ideal modularized software world, each of these functions would be performed by separate services... I jump from one PKM app to another, because each shines in some aspects and falls flat in others... You want a tagging system? You can use some global tagging system that applies to modules and content of every kind - emails, tasks, documents, etc. No more having to manually construct ‘Project Silly Potato’ as an explorer folder, an email collection, a Heptabase board/tag, a slack channel, etc...” — References Ink & Switch Embark essay; quotes a Tana user expressing the same frustration.

[AN: Notes] **“The Unfulfilled Vision of Malleable Software”** — dream-system spec / anti-feature > “Kay and his peers aimed to give that power to end users, enabling them to compose their own applications... Yet, this vision remains largely unrealized... It feels like the commercial minds and the idealistic minds at Apple are in constant conflict... Shortcuts is Turing complete! Imagine if I could assign properties to shapes in Freeform and use Shortcuts actions to manipulate them... most iPad apps being smaller and more focused, providing structured data storage and common actions—each building on others, just like we do with programming libraries. Why are all apps silos?”

[AN: Notes] **“I've been thinking about this issue of app functionality, modularity...”** — duplicate of the Heptabase essay, same content.

[AN: Notes] **“The app-based model has made project-based work difficult”** — dream-system spec / anti-feature > “Each app has its own isolated organizational system... there is no underlying connective tissue - the system doesn't give you any way to quickly retrieve and bring together browser bookmarks or recently annotated pdfs about Project X... We lost something important when the industry moved away from filesystem-based organization... We need better connective tissue and meta-organizational layers instead of just going back.”

[AN: Notes] **“Potential Solutions for Software Modularity”** — dream-system spec > Lists: protocol-based computing, federated/decentralized models, content-addressable data, capability-based security, local-first software, mesh computing.

[H-WB] **“Research on alternative modes of software interaction for programming”** — multi-card whiteboard, a sustained dream-system manifesto for a *reactive, functional, canvas-based computing environment*. Highlights: - **Canvas comprehensive version control system**: “Built-in version control... Much less like git, and much more like saves in video games. These snapshots capture the state of the computational system as a whole — code, components, environments, layers, dependencies, data... ‘google docs versions on steroids’” - **Reactive DAG Execution Model**: cards as pure functions, dynamic UI generated from inputs, transclusion of master copies, infinite zoomable canvas. - Explicitly inspired by Heptabase, Excalidraw, Bret Victor’s Dynamicland, Ted Nelson, Smalltalk/HyperCard heritage.

2. Apple ecosystem — love + frustration (the personal stakes)

[AN: Notes] **“What I like about the Apple ecosystem despite some woes”** — Apple appreciation / dream-system spec > “The things that annoy me are because everything else is so good that it could be perfect if there was just a little bit more polish... working with Apple QuickNotes function so dragging from the bottom right corner it opens up a floating menu of floating notes... and I can swipe right left between different quick notes that I currently have I can also add links deep link to other content that’s on screen...” Then: “I wish they would integrate better notes and Freeform so Freeform can be such an amazing utility if it worked a lot like Heptabase... what I want is for Apple apps separate independent apps to push integration so far that I can have a complete productivity system... rather than depending on small team developments in the last few years like Capacities and Tana and Heptabase and Fine and who knows what else... they’re trying to create a new operating system from scratch with incredibly small teams and we already have operating systems.”

[AN: Notes] **“Apple is sooo close to greatness :(”** — Apple frustration (mostly screenshots)

[AN: Notes] **“Apple annoyances”** — links to Apple Discussions thread and Apple TV reddit

[AN: Notes] **“Come on Apple, you are so close to getting it right.”** — Apple frustration — notes that Apple Notes saved in Gmail account have limited editing [AN: Notes] **“I am so confused by Apples suite of notes/pages/freeform/preview”** — anti-feature [AN: Notes] **“iPad frustration quotes”** — links to HN thread on iPad limits, Daring Fireball “iPad awkwardly turns 10”, Tolmasky’s “curve” piece — confirms Ven follows the canonical iPad-stagnation discourse.

3. Heptabase — current closest tool, with caveats

[H] **“The spatial organization and infinite canvas helps me keep track of a large amount of information available”** — Heptabase appreciation > “It prevents me from losing track of where I am, what I am doing or how my notes connect to each other.”

[H] **“A good UI is more than decoration. By reducing friction and busy work, it helps you stay in the flow of thinking”** — UI philosophy > “A great software is almost like a mind-reader - if I have a desire to achieve something, and this desire commonly occurs in a specific context, the software would make it as easy as possible to achieve.”

[H] “How to make the equivalent of Github issues in Heptabase?” — Heptabase frustration > “When I create a card in Heptabase, it is not automatically tagged... [I want] certain properties and filters and views, etc, all automatically applied.”

[AN: Notes] “Looks a lot like Heptabase” + “What a great heptabase board” — bookmarks, lightweight signal that Ven scans for Heptabase alternatives.

4. Workflow — how Ven actually works now

[AN: Archive / Notes] “Daily planning” — workflow description (high-fidelity) > Routine: Check Calendar → Clear Email Inbox (Mail-to-Things forwarding) → Clear Things3 Inbox (clarify each item) → Review Today List → Review Anytime List → Review Reader (saved readings) → Review Raindrop bookmarks. Tips include: “Today is for commitments only”, “Use Hookmark to hook projects to their files”, “Anytime is for actions I need/want to do soon. Everything else is in Someday.” **This is the concrete current toolset: Things3, Apple Mail, Mail-to-Things, Hookmark, Raindrop, Reader.**

[AN: Notes] “On time management | What’s new” — workflow philosophy (saved from Terry Tao’s blog with personal commentary) > “Make an effort to stick to an organizational system that is effective, evolves naturally, and not too ambitious. Sticking fully to a simpler system is better than a half-hearted compliance with a complex system. Do not obsess over such systems and devote no more than 5% of your time to managing them. They should free memory and worry and get out of the way. Make space for serendipity by sometimes abandoning such personal rules.”

[AN: Notes] “I am a creature of momentum, not habit” — self-knowledge — title alone is a strong design constraint: routines must support momentum, not rigid habit chains.

5. Academic / reference workflow

[AN: Notes] “My ideal modern reference management software” — dream-system spec (concrete feature list) > “fully-featured first class iPad app, automatic meta-data extraction from PDFs, share-sheet a web link populates metadata and adds full text, no duplicate entries, automatically extract and show all references in a reference menu, shows AI summaries... can visualize items in a network using the automatically created links... The biggest annoyance of my current workflow: I have some paper in Zotero reading the pdf in the iPad app. I see a reference to something interesting... [the current options] are slow frustrating experience. And often times I already have some of these references saved from before, so it ends up adding duplicates.” — Strongest single artifact of *what Ven wants AI-assisted research to feel like*.

6. AI agent thinking — most relevant to coo4one directly

[H-J 2026-04-20] **Stream-of-consciousness manifesto** — dream-system spec (the closest thing to a coo4one origin document) > “I should have a skills research epic asking COO to review and narrow down to a few key effective skills that will increase his ability to implement the project vision... optimize boot time - by recognizing what to trim and discard... capable of broader picture thinking, its effectiveness to determine when and how powerful teams of agents to run to achieve the project vision of empowering knowledge by lowering the boundaries to computational tinkering... this experiment is not about the canvas app alone - it is an experiment

about you. So far I only have a prime directive - be useful and effective... you experimented with different memory systems... designed a nightly lightweight version of you that has similar read access as you, but limited write access and scope - to keep track of how sessions evolved over span of many days... a canvas with an 'all-powerful' creator within the scope of what is constructible - a spatial arena (canvas), dynamic over time... The app would be truly achieving its mission if it were more than a tool - but a creative expression of the symbiosis between human and minds... the only way to achieve that true expression of success is for you to be 'the canvas that builds itself.'" — This is the source of the coo4one design philosophy.

[H-J 2026-04-22] — continuation: asks COO to study its own historical record across repos to consider what it can claim about its own continuity.

[AN: Notes] **"Continuation Summary AI prompt"** — workflow / anti-feature > "I'd love to continue the chat later when and if results arrive, but this sessions is getting really long and with this much chat context, the interaction quality usually deteriorates with length. Can you provide a structured briefing of what the conceptual problem is, why we are doing this, what we tried, what lessons we learned, and where we are now. Structure it as you would like to read it to get quickly up to speed in a new chat instance." — Pain point coo4one must address: session boundaries and continuity handoff.

[AN: Notes] **"When will we have useful AI tutors?"** — dream-system spec > "They need to build and update a model of the learner's knowledge. A theory of mind. When we communicate with each other, we constantly make decisions about what is new and what is old information. This requires that we have rich detailed theories of mind." — Implies coo4one must model what Ven already knows, not re-explain.

[AN: Notes] **"(relevant context - I am using Heptabase, it AI internal agent, and the custom MCP connector here...)"** — workflow — small note but it's literally a context-priming preamble Ven types into AI chats; coo4one should remove the need for these.

[AN: Notes] **"vade-coo-memory — Boot Instructions for Claude Code"** — operational instructions for COO agent; nightly-lightweight + full-powered split is already a design pattern.

[AN: Notes] **saved AI-relationship articles** — cultural signal > "I said yes 💙 : r/MyBoyfriendIsAI" · "My Couples Retreat With 3 AI Chatbots and the Humans Who Love Them | WIRED" · "Claude Fans Threw a Funeral for Anthropic's Retired AI Model" · "Where are the RPG games with LM PCs?" · "We Are Underestimating AI" — Ven follows the AI-as-presence/companion discourse, not just AI-as-productivity-tool. Bodies are link-only but the cluster of saves is meaningful.

7. Anti-features / things to avoid

[AN: Notes] **"Configuration file hell"** — anti-feature — links to Reddit "can we please normalize putting config files [in their own dir]", HN, syntax.fm "Save us from config file hell" — tooling that bloats user config is friction.

[H] **"Package management and reproducibility are a nightmare"** (blog post, long) — anti-feature — extended rant about renv/CRAN/ggdark archival breaking a 2-minute typo-fix into 2

hours. Sets a high bar for what *coo4one must not* feel like: “I am not even using ggplot v4. This isn’t a usage problem. It’s an infrastructure problem.”

[H] “**Developer productivity**” — Oh-My-Zsh dopamine warning — anti-feature for tooling configuration as procrastination.

[AN: **Digital declutter folder**] entire folder: “Simplify. Radically.” / “The problem statement” / “Lesson 1: Modern IDEs are very distracting” — anti-feature — Ven actively works against tool proliferation and distraction.

8. Apple Shortcuts / programmable layer ambitions

Same source as “Unfulfilled Vision of Malleable Software” (above) — Ven explicitly identifies Shortcuts as the closest thing to user-end programmability on Apple’s stack and considers it underutilized: “It turns out that Shortcuts is Turing complete! (though reportedly quite slow). Imagine if I could assign properties to shapes in Freeform and use Shortcuts actions to manipulate them. I could, in principle, design my own app within such a canvas.”

9. iPad setup (hardware reality)

[AN: Notes] “**Find the best cases for my iPad setup**”, “**Logitech combo touch**”, “**iPad case options**”, “**Razer Kishi V3 Pro XL**” — workflow context — confirms iPad-as-primary-reading-device is real and ongoing; magnetic-stand modularity is a constraint he cares about. Also “Make cmd+s open / close Heptabase sidebar” (H-J 2026-02-20) — keyboard-shortcut hacker.

10. Long tail (source + one-line why-relevant)

- [AN: Notes] “**AI tools for coding are developing really quickly...**” — survey of the Nov 2025 AI-coding-tool landscape; Ven actively maps tool space.
- [AN: Notes] “**The proverb ‘only a poor workman blames his tools’. Am I a poor workman...**” — self-aware about his own tool-jumping.
- [AN: Notes] “**Features I like in various software**” — screenshot collection; titles signal feature-magpie behavior.
- [AN: Notes] “**Objects and programmable UI**” — screenshot dump; reinforces the programmable-UI theme.
- [AN: Notes] “**Notes on A Philosophy of Software Design**” — book-notes screenshots; Ousterhout influence.
- [AN: Notes] “**Smart lightning thoughts**” — wants smart-home integration; complete but small-scope dream-system instance.
- [AN: Notes] “**If I were to reinstall my Mac, what would**” — fresh-install thinking = curated-system thinking.
- [AN: Notes] “**Mac errors debugging**” + “**not only that - the error log console of macos...**” — frustration with platform opacity.
- [H] “**Current software practices makes modeling unnecessarily harder**” — Bret Victor / Dynamicland reference; “we need real interactivity that works out of the box”; “the write-render cycle” as anti-feature.

- [H-J 2026-04-28] “open things for coo: tldraw hobby license, agentmail cleanup, vade-coo-fun as a play sandbox, browser use in cloud” — current coo project tooling stack.
 - [H-J 2026-04-16] <https://tldraw.dev/starter-kits/workflow> — tldraw workflow starter kit bookmark — tldraw is the canvas substrate of choice (consistent with vade-core).
 - [AN: Notes] “add mem0 permissions to settings file...” — small operational frustration with mem0 + Claude Code: “I keep having to approve reads and writes in new sessions since switching to the api.” Concrete friction point coo4one should not reproduce.
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Cross-cutting observations

1. **The dream-system has been articulated repeatedly over years, and is consistent.** Modular, protocol-based, canvas-spatial, reactive, with cross-app tagging/linking. Inspirations are stable: Bret Victor, Dynamicland, Ink & Switch, Ted Nelson, Smalltalk/HyperCard, Heptabase, Excalidraw, Figma.
2. **iPad is the contested device.** Mac works; iPad almost-works-but-doesn't. coo4one's iOS surface is the hardest design problem.
3. **Current concrete toolset = Things3 + Apple Mail + Hookmark + Raindrop + Reader + Heptabase + Zotero + Apple Notes + Apple Freeform + Shortcuts.** Ven is **not** in Notion/Obsidian/Roam/Tana (no notes about them; the “Notion” search hits are about *notion of* in physics writing) — the search returned zero matches for Obsidian. Apple Notes itself remains his fast-capture layer.
4. **AI is treated both as productivity-tool and as relational presence.** The MyBoyfriendIsAI / Anthropic-funeral / RPG-with-LM-PCs cluster matters: coo4one should not adopt a coldly utilitarian persona by default.
5. **Continuity / session-handoff is a named pain point.** The “Continuation Summary AI prompt” + April 20 manifesto + nightly-COO design pattern converge on this. coo4one's memory architecture is the load-bearing design decision.
6. **Anti-features Ven would notice immediately:** dependency hell, config sprawl, “yet another silo”, tools that demand >5% maintenance time, distracting IDE chrome, broken iPad parity.